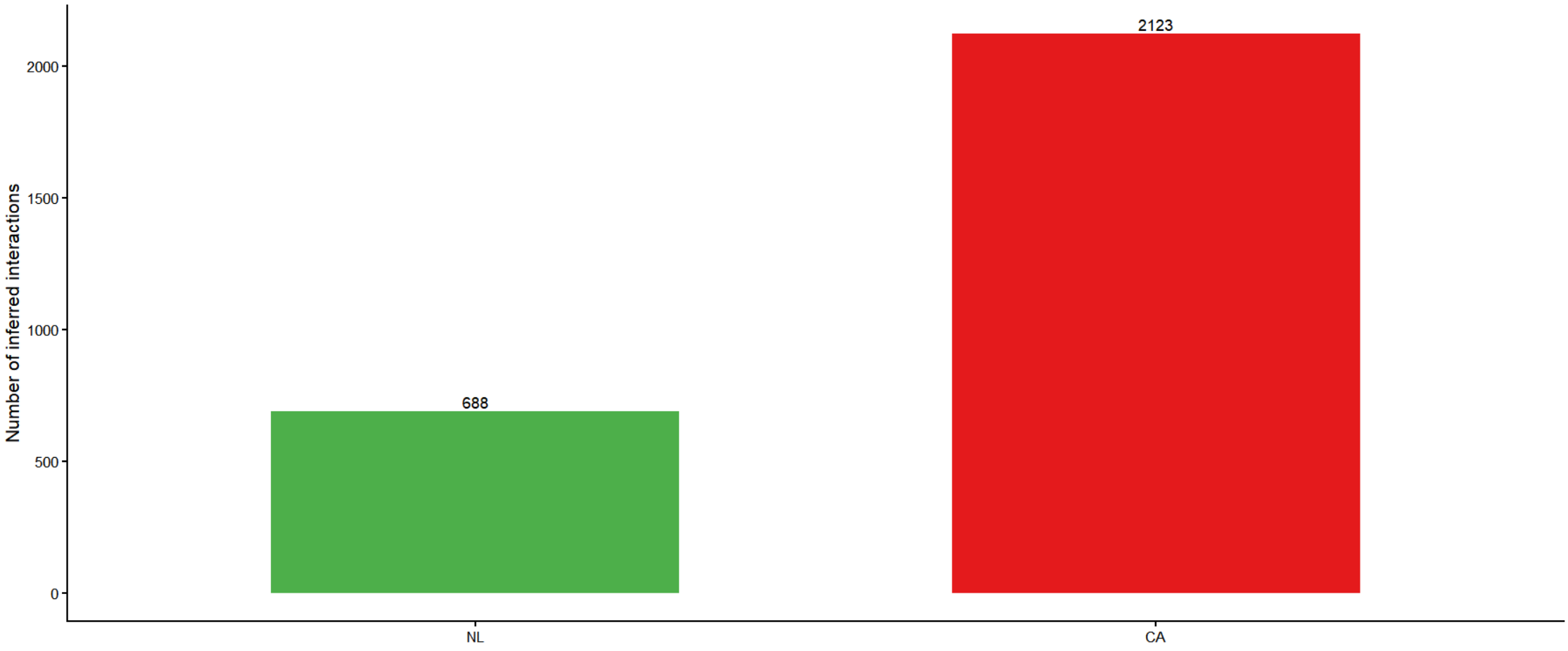
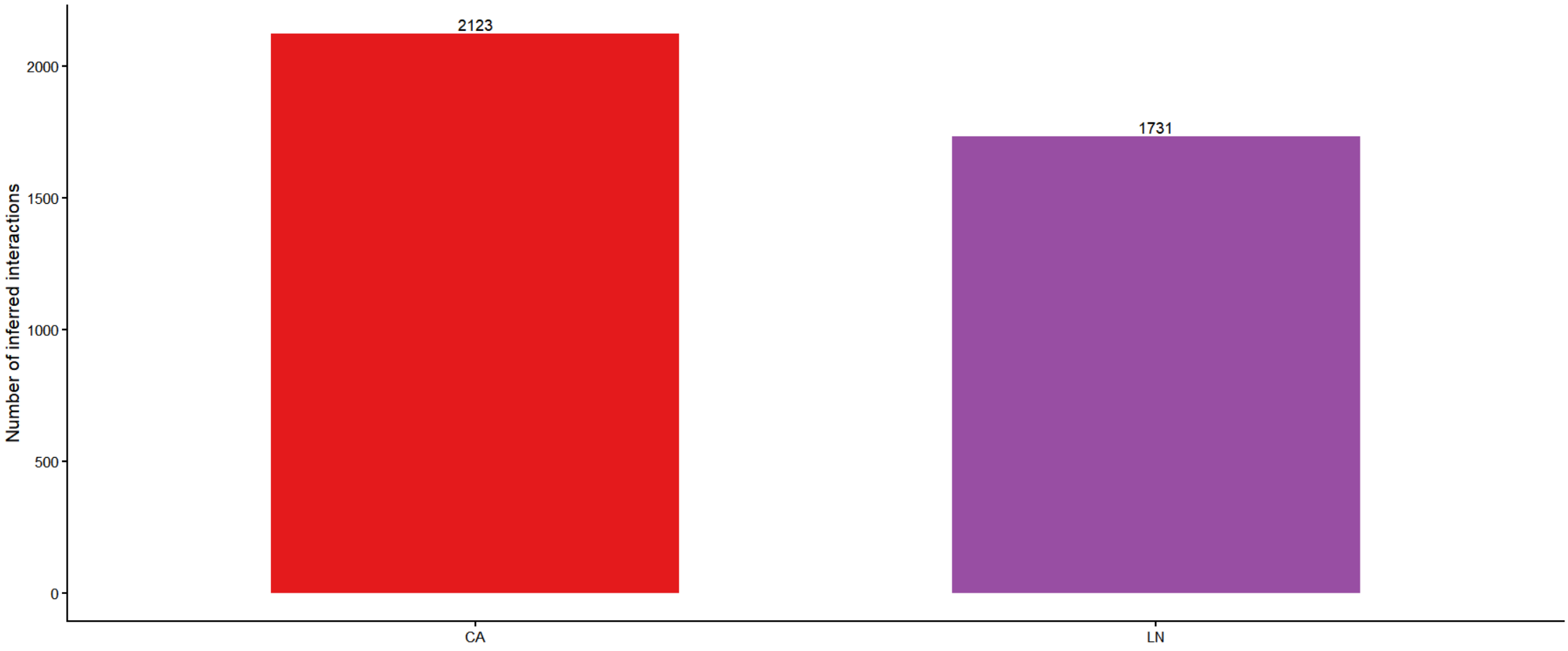
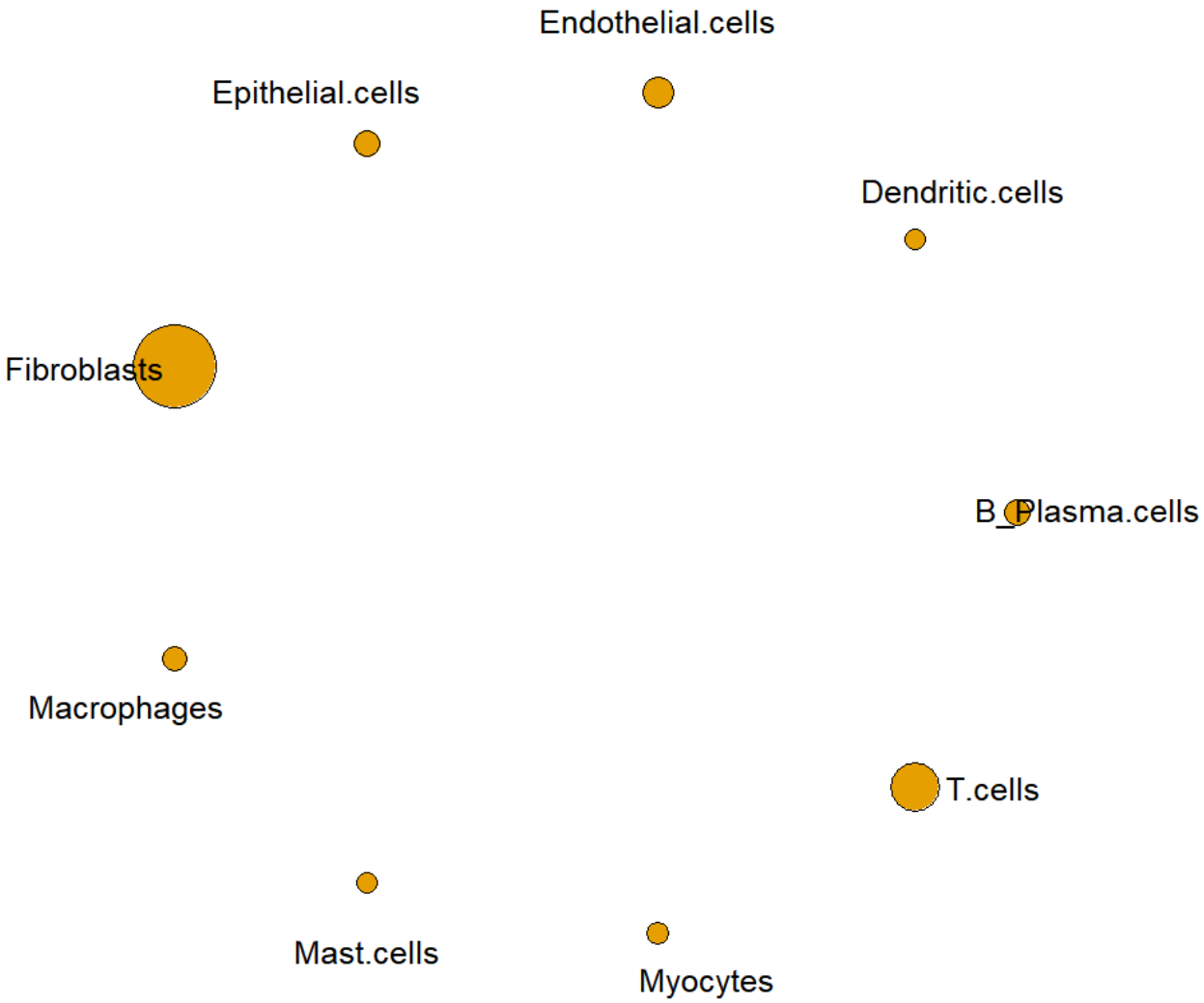
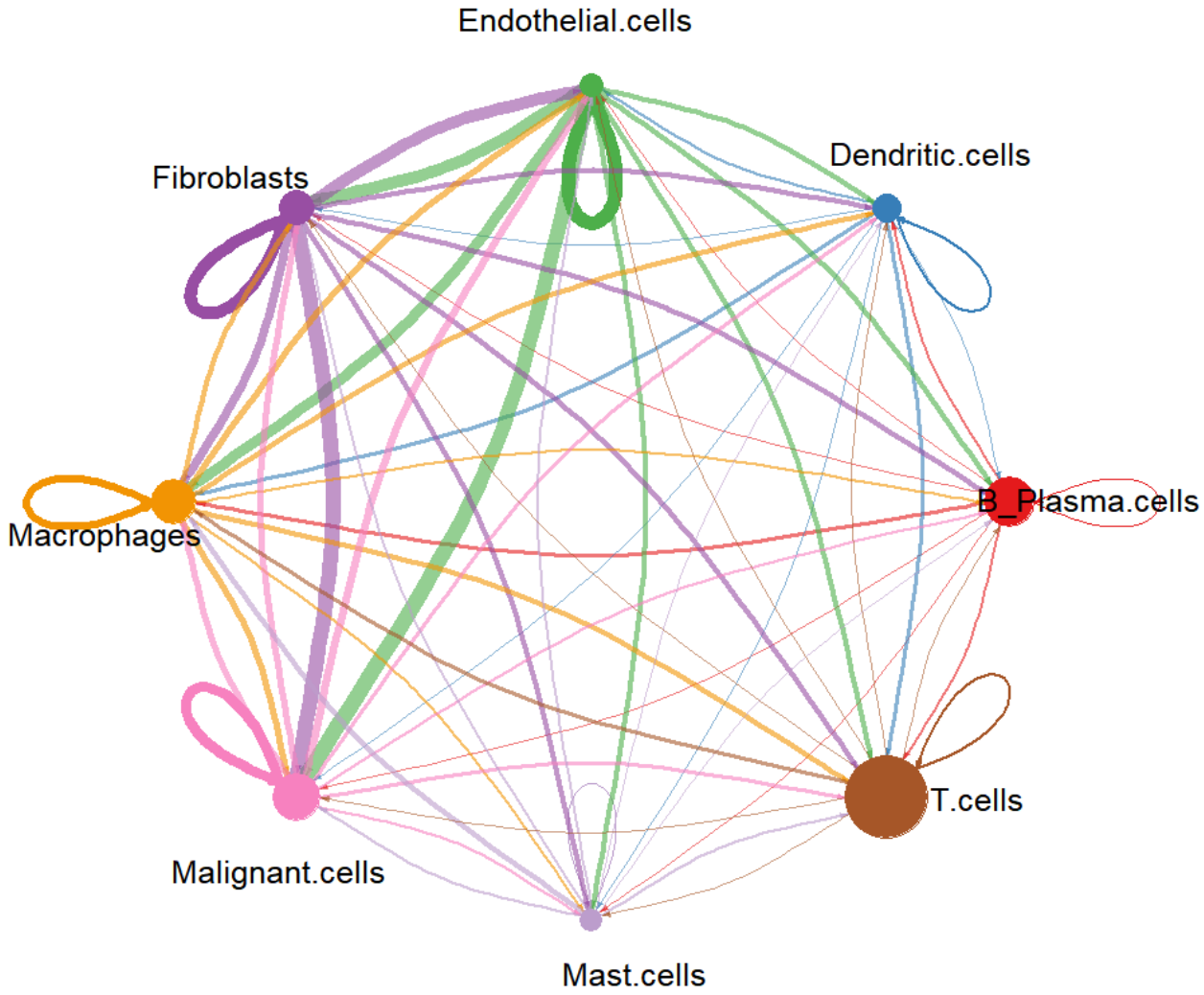
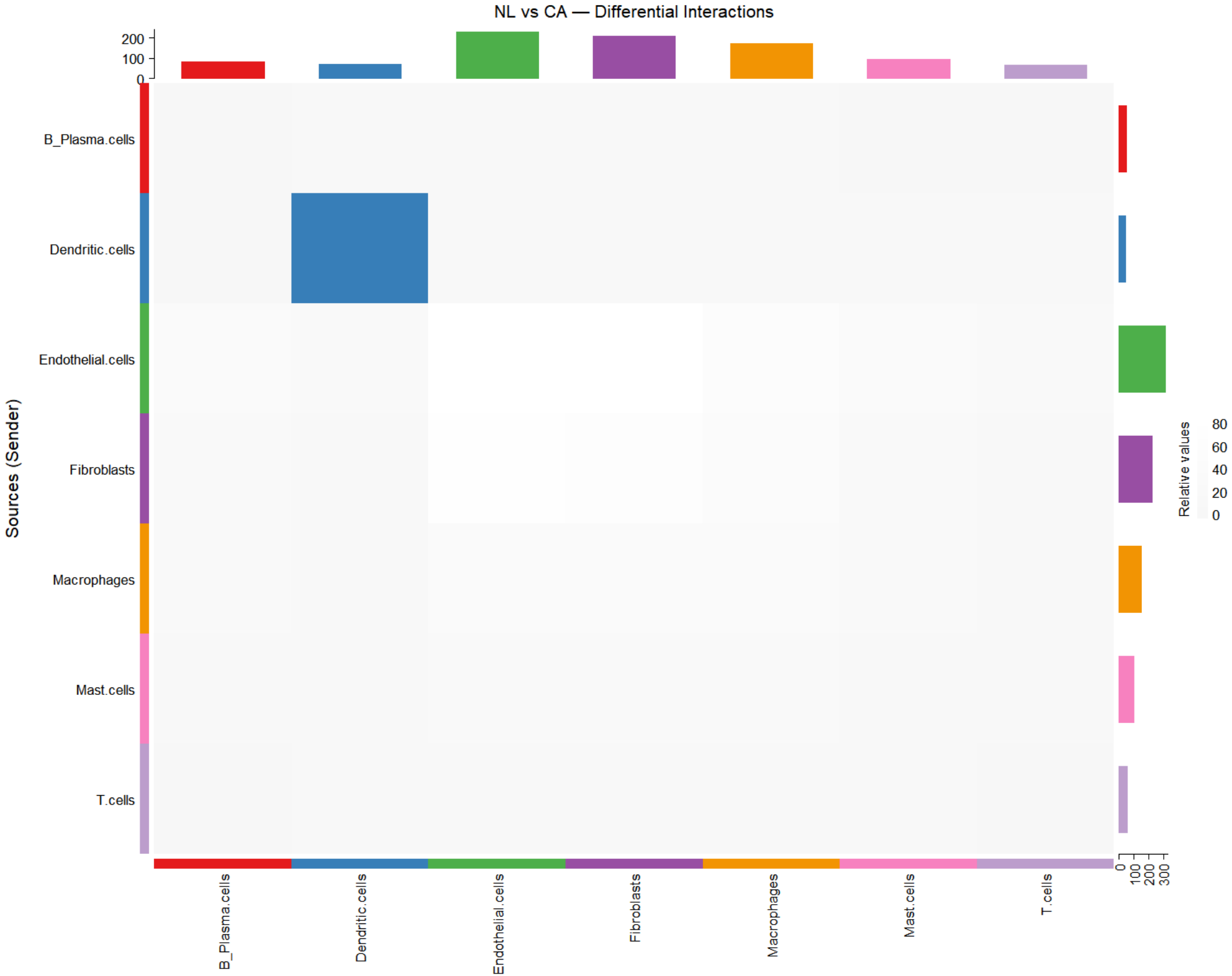
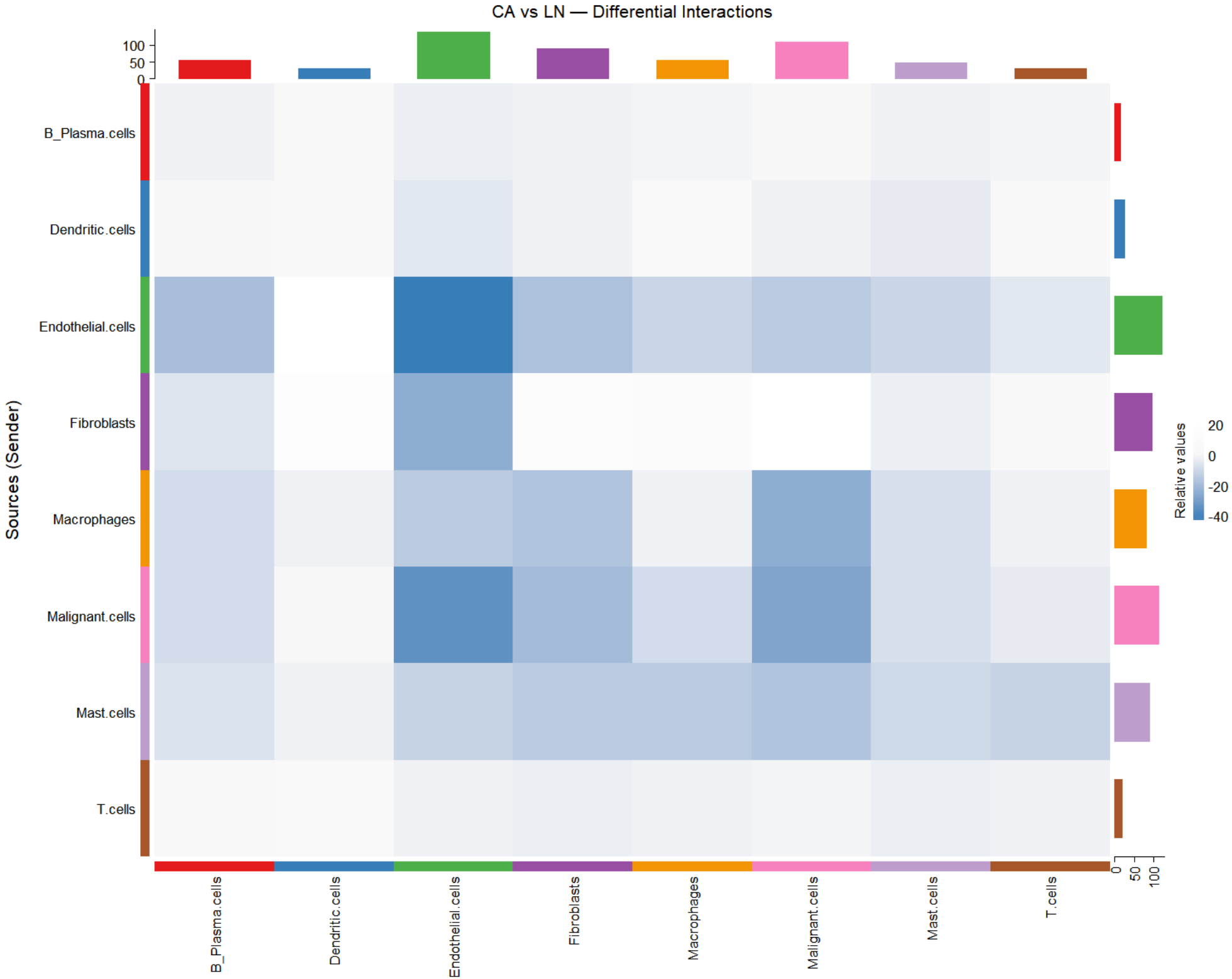

Number of inferred interactions
0
500
1000
1500
2000
688
2123
NL
CA

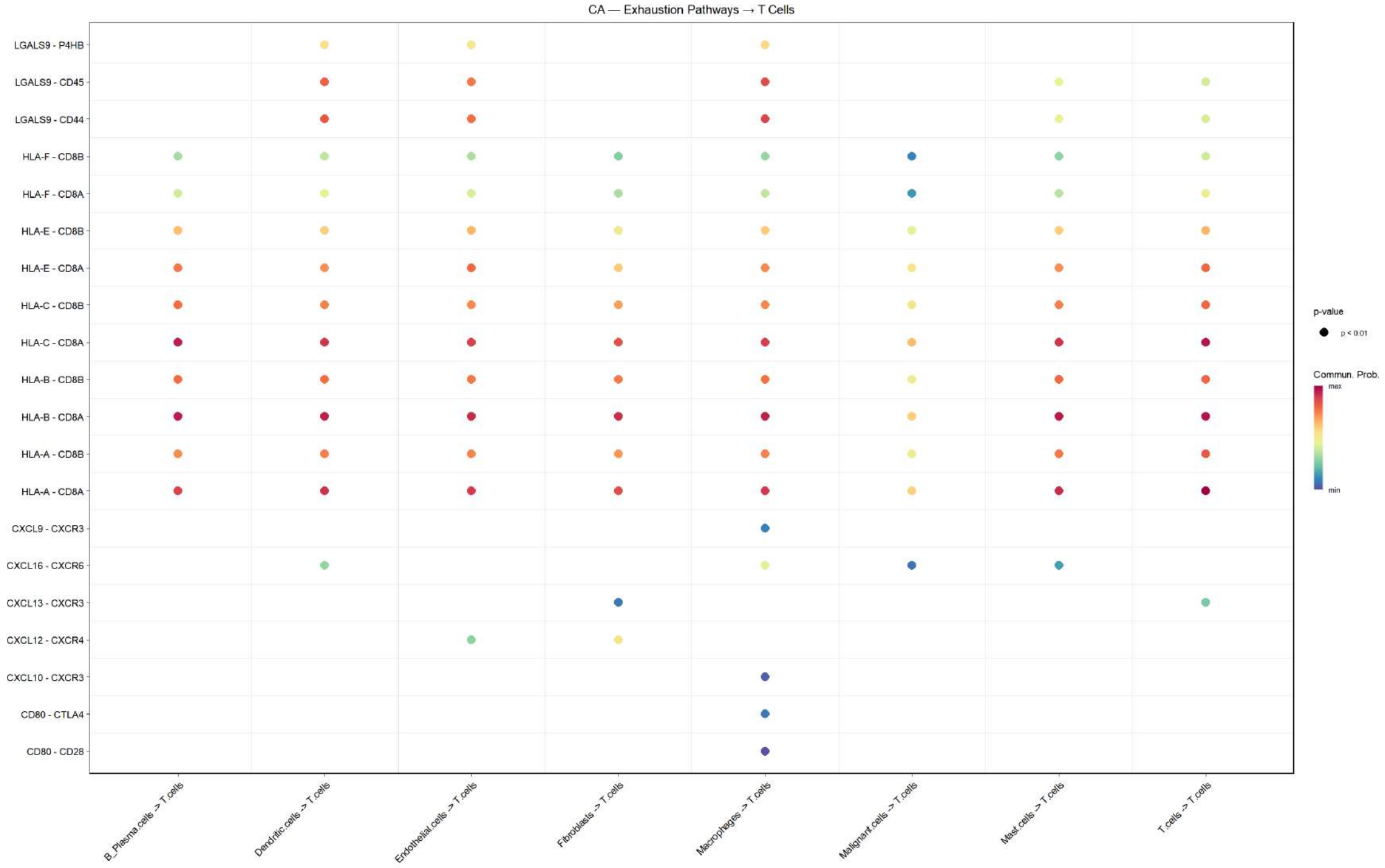

2123
1731
Number of inferred interactions
2000
1500
1000
500
0
CA
LN

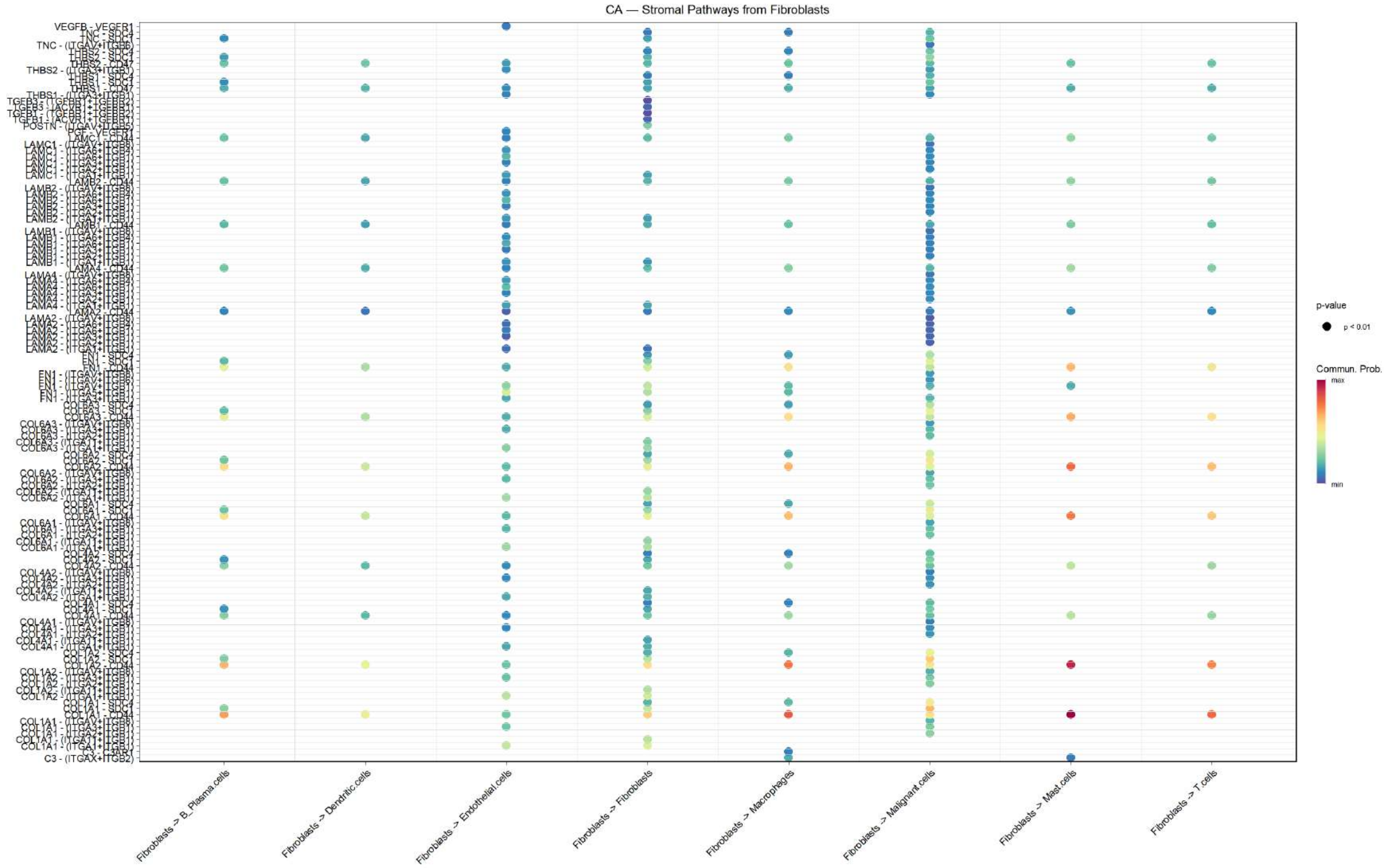

Endothelial.cells
Epithelial.cells
Dendritic.cells
Fibroblasts
B_Plasma.cells
Macrophages
T.cells
Mast.cells
Myocytes

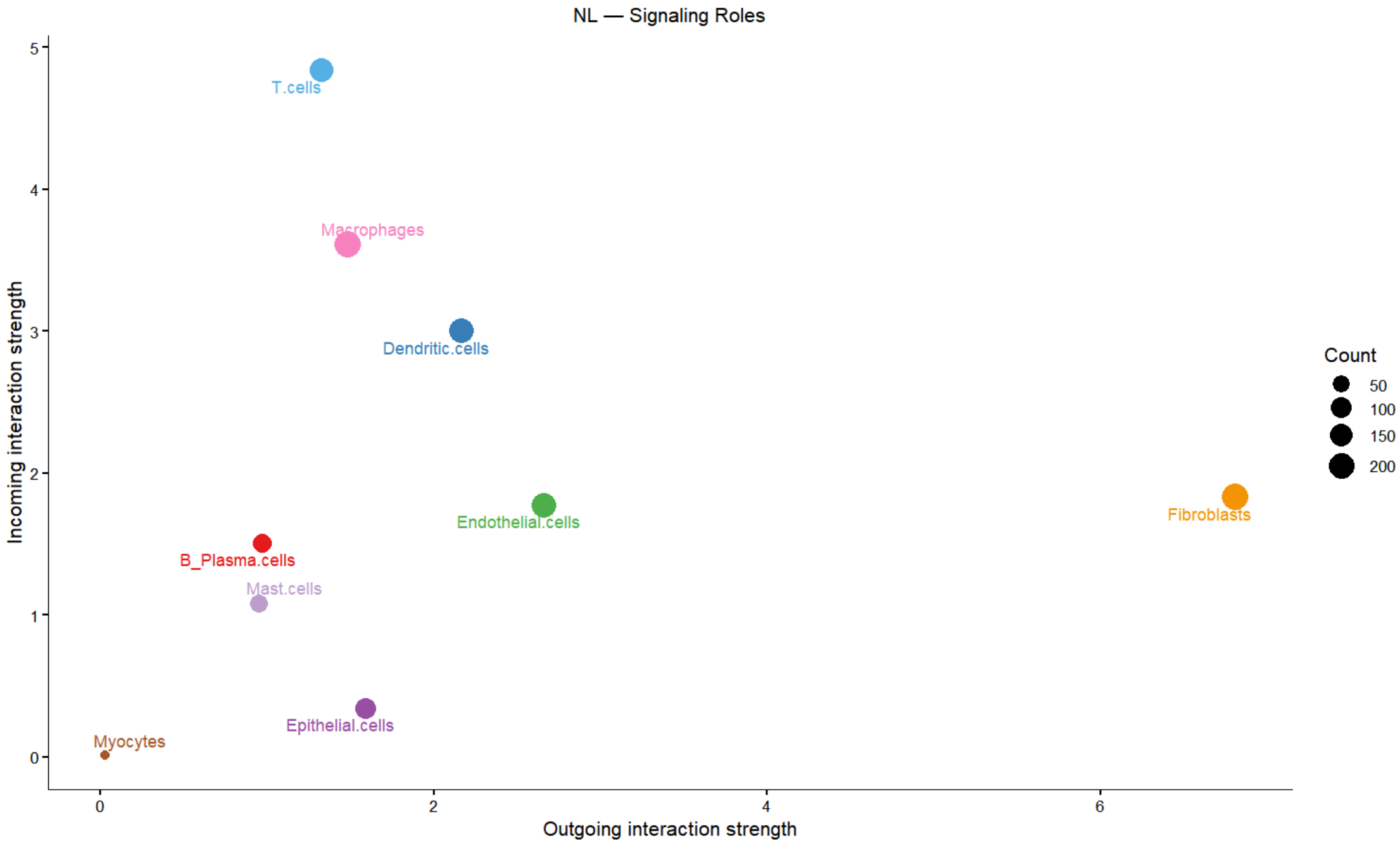

Endothelial.cells
Dendritic.cells
Fibroblasts
B_Plasma.cells
Macrophages
T.cells
Malignant.cells
Mast.cells

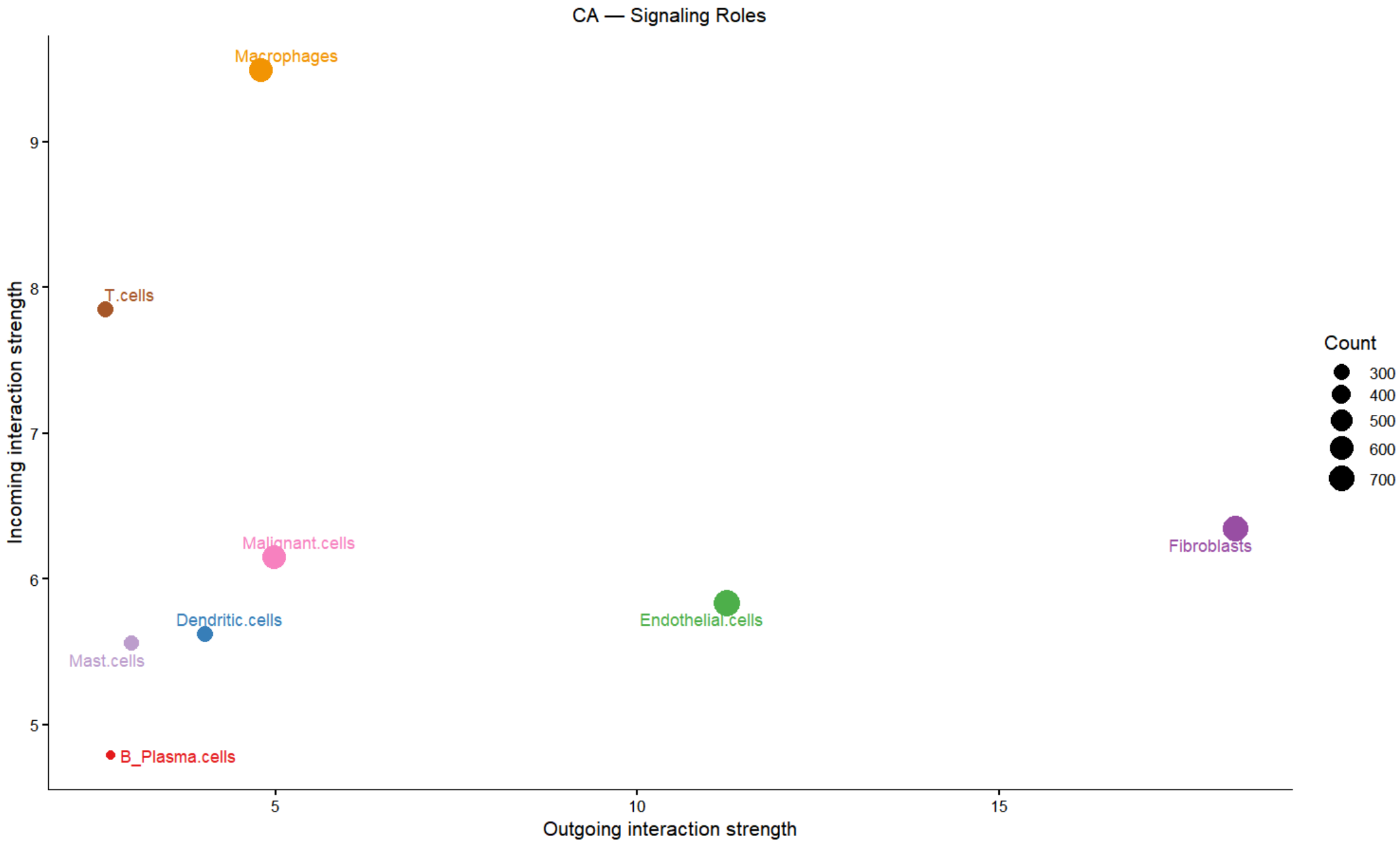

NL vs CA — Differential Interactions
200
100
0
Sources (Sender)
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Mast.cells
T.cells
Relative values
80
60
40
20
0
0
100
200
300
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Mast.cells
T.cells

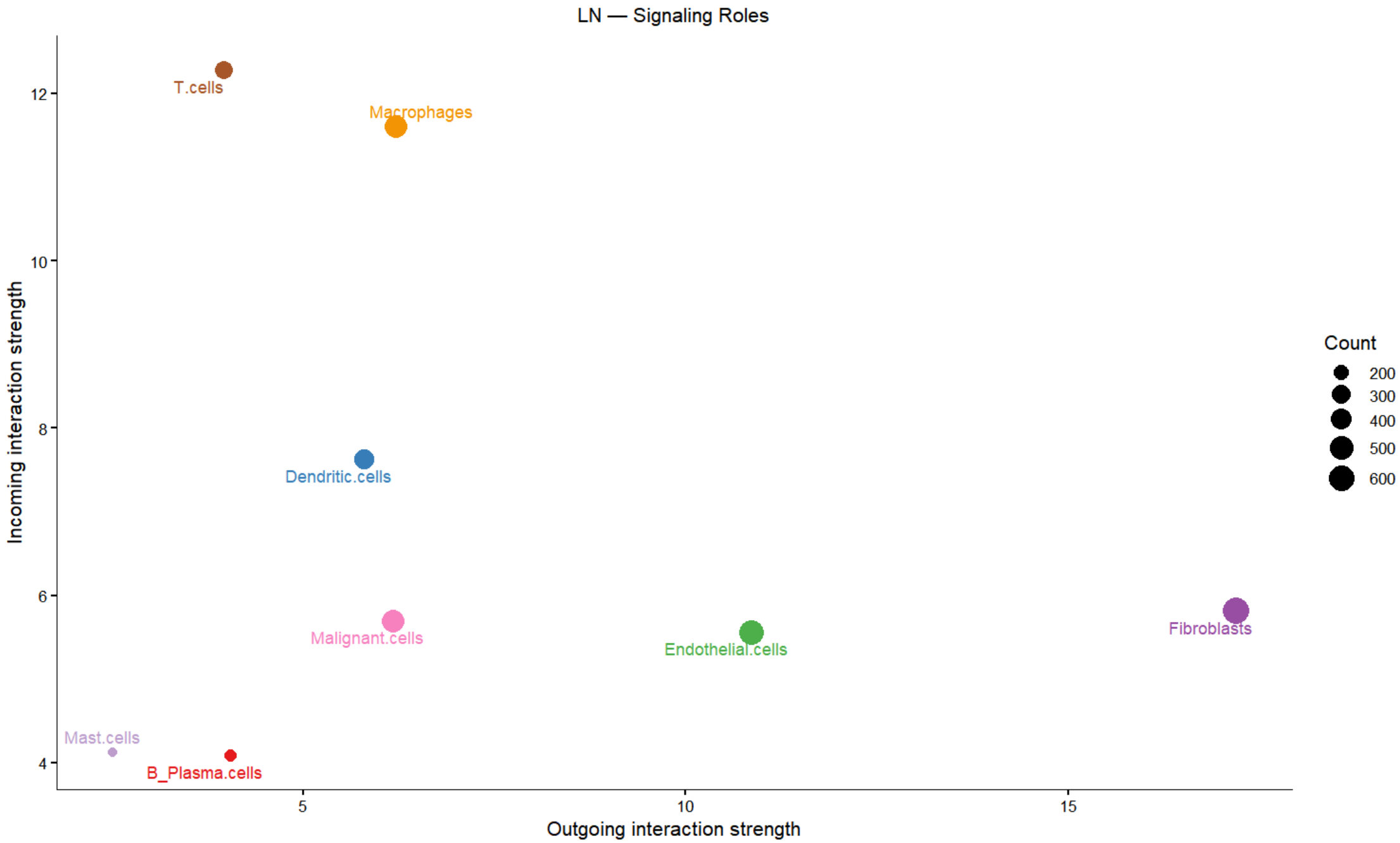

CA vs LN — Differential Interactions
Sources (Sender)
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Malignant.cells
Mast.cells
T.cells
Relative values
20
0
-20
-40
100
50
0

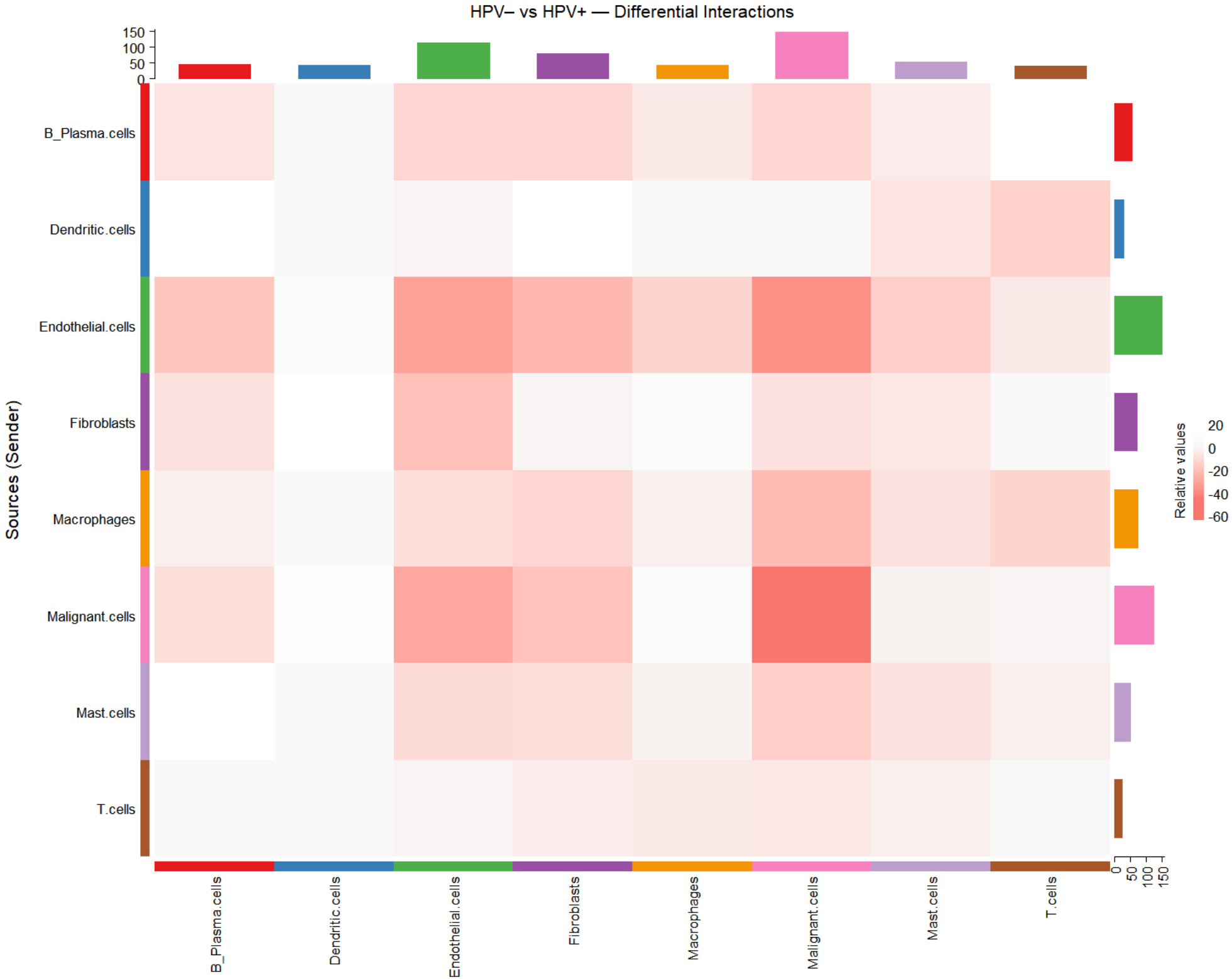

CA — Exhaustion Pathways → T Cells
LGALS9 - P4HB
LGALS9 - CD45
LGALS9 - CD44
HLA-F - CD8B
HLA-F - CD8A
HLA-E - CD8B
HLA-E - CD8A
HLA-C - CD8B
HLA-C - CD8A
HLA-B - CD8B
HLA-B - CD8A
HLA-A - CD8B
HLA-A - CD8A
CXCL9 - CXCR3
CXCL16 - CXCR6
CXCL13 - CXCR3
CXCL12 - CXCR4
CXCL10 - CXCR3
CD80 - CTLA4
CD80 - CD28
B_Plasma.cells -> T.cells
Dendritic.cells -> T.cells
Endothelial.cells -> T.cells
Fibroblasts -> T.cells
Macrophages -> T.cells
Malignant.cells -> T.cells
Mast.cells -> T.cells
T.cells -> T.cells
p-value
p < 0.01
Commun. Prob.
max
min


CA — Stromal Pathways from Fibroblasts
VEGFB - VEGFR1
TNC - SDC4
TNC - SDC1
TNC - (ITGAV+ITGB6)
THBS2 - SDC4
THBS2 - SDC1
THBS2 - CD47
THBS2 - (ITGA3+ITGB1)
THBS1 - SDC4
THBS1 - SDC1
THBS1 - CD47
THBS1 - (ITGA3+ITGB1)
TGFB3 - (TGFBR1+TGFBR2)
TGFB3 - (ACVR1+TGFBR2)
TGFB1 - (TGFBR1+TGFBR2)
TGFB1 - (ACVR1+TGFBR2)
POSTN - (ITGAV+ITGB5)
PGF - VEGFR1
LAMC1 - CD44
LAMC1 - (ITGAV+ITGB6)
LAMC1 - (ITGA6+ITGB1)
LAMC1 - (ITGA6+ITGB4)
LAMC1 - (ITGA3+ITGB1)
LAMC1 - (ITGA2+ITGB1)
LAMC1 - (ITGA1+ITGB1)
LAMB2 - CD44
LAMB2 - (ITGAV+ITGB6)
LAMB2 - (ITGA6+ITGB1)
LAMB2 - (ITGA6+ITGB4)
LAMB2 - (ITGA3+ITGB1)
LAMB2 - (ITGA2+ITGB1)
LAMB2 - (ITGA1+ITGB1)
LAMB1 - CD44
LAMB1 - (ITGAV+ITGB6)
LAMB1 - (ITGA6+ITGB1)
LAMB1 - (ITGA6+ITGB4)
LAMB1 - (ITGA3+ITGB1)
LAMB1 - (ITGA2+ITGB1)
LAMB1 - (ITGA1+ITGB1)
LAMA4 - CD44
LAMA4 - (ITGAV+ITGB6)
LAMA4 - (ITGA6+ITGB1)
LAMA4 - (ITGA6+ITGB4)
LAMA4 - (ITGA3+ITGB1)
LAMA4 - (ITGA2+ITGB1)
LAMA4 - (ITGA1+ITGB1)
LAMA2 - CD44
LAMA2 - (ITGAV+ITGB6)
LAMA2 - (ITGA6+ITGB1)
LAMA2 - (ITGA6+ITGB4)
LAMA2 - (ITGA3+ITGB1)
LAMA2 - (ITGA2+ITGB1)
LAMA2 - (ITGA1+ITGB1)
FN1 - SDC4
FN1 - SDC1
FN1 - CD44
FN1 - (ITGAV+ITGB1)
FN1 - (ITGAV+ITGB6)
FN1 - (ITGAV+ITGB5)
FN1 - (ITGA5+ITGB1)
FN1 - (ITGA4+ITGB1)
COL6A3 - SDC4
COL6A3 - SDC1
COL6A3 - CD44
COL6A3 - (ITGAV+ITGB6)
COL6A3 - (ITGA3+ITGB1)
COL6A3 - (ITGA2+ITGB1)
COL6A3 - (ITGA1+ITGB1)
COL6A3 - (ITGA1+ITGB1)
COL6A2 - SDC4
COL6A2 - SDC1
COL6A2 - CD44
COL6A2 - (ITGAV+ITGB6)
COL6A2 - (ITGA3+ITGB1)
COL6A2 - (ITGA2+ITGB1)
COL6A2 - (ITGA1+ITGB1)
COL6A2 - (ITGA1+ITGB1)
COL6A1 - SDC4
COL6A1 - SDC1
COL6A1 - CD44
COL6A1 - (ITGAV+ITGB6)
COL6A1 - (ITGA3+ITGB1)
COL6A1 - (ITGA2+ITGB1)
COL6A1 - (ITGA1+ITGB1)
COL6A1 - (ITGA1+ITGB1)
COL4A2 - SDC4
COL4A2 - SDC1
COL4A2 - CD44
COL4A2 - (ITGAV+ITGB6)
COL4A2 - (ITGA3+ITGB1)
COL4A2 - (ITGA2+ITGB1)
COL4A2 - (ITGA1+ITGB1)
COL4A2 - (ITGA1+ITGB1)
COL4A1 - SDC4
COL4A1 - SDC1
COL4A1 - CD44
COL4A1 - (ITGAV+ITGB6)
COL4A1 - (ITGA3+ITGB1)
COL4A1 - (ITGA2+ITGB1)
COL4A1 - (ITGA1+ITGB1)
COL4A1 - (ITGA1+ITGB1)
COL1A2 - SDC4
COL1A2 - SDC1
COL1A2 - CD44
COL1A2 - (ITGAV+ITGB6)
COL1A2 - (ITGA3+ITGB1)
COL1A2 - (ITGA2+ITGB1)
COL1A2 - (ITGA1+ITGB1)
COL1A2 - (ITGA1+ITGB1)
COL1A1 - SDC4
COL1A1 - SDC1
COL1A1 - CD44
COL1A1 - (ITGAV+ITGB6)
COL1A1 - (ITGA3+ITGB1)
COL1A1 - (ITGA2+ITGB1)
COL1A1 - (ITGA1+ITGB1)
COL1A1 - (ITGA1+ITGB1)
C3 - IAR
C3 - (ITGAX+ITGB2)
Fibroblasts -> B_Plasma.cells
Fibroblasts -> Dendritic.cells
Fibroblasts -> Endothelial.cells
Fibroblasts -> Fibroblasts
Fibroblasts -> Macrophages
Fibroblasts -> Malignant.cells
Fibroblasts -> Mast.cells
Fibroblasts -> T.cells
p-value
p < 0.01
Commun. Prob.
max
min

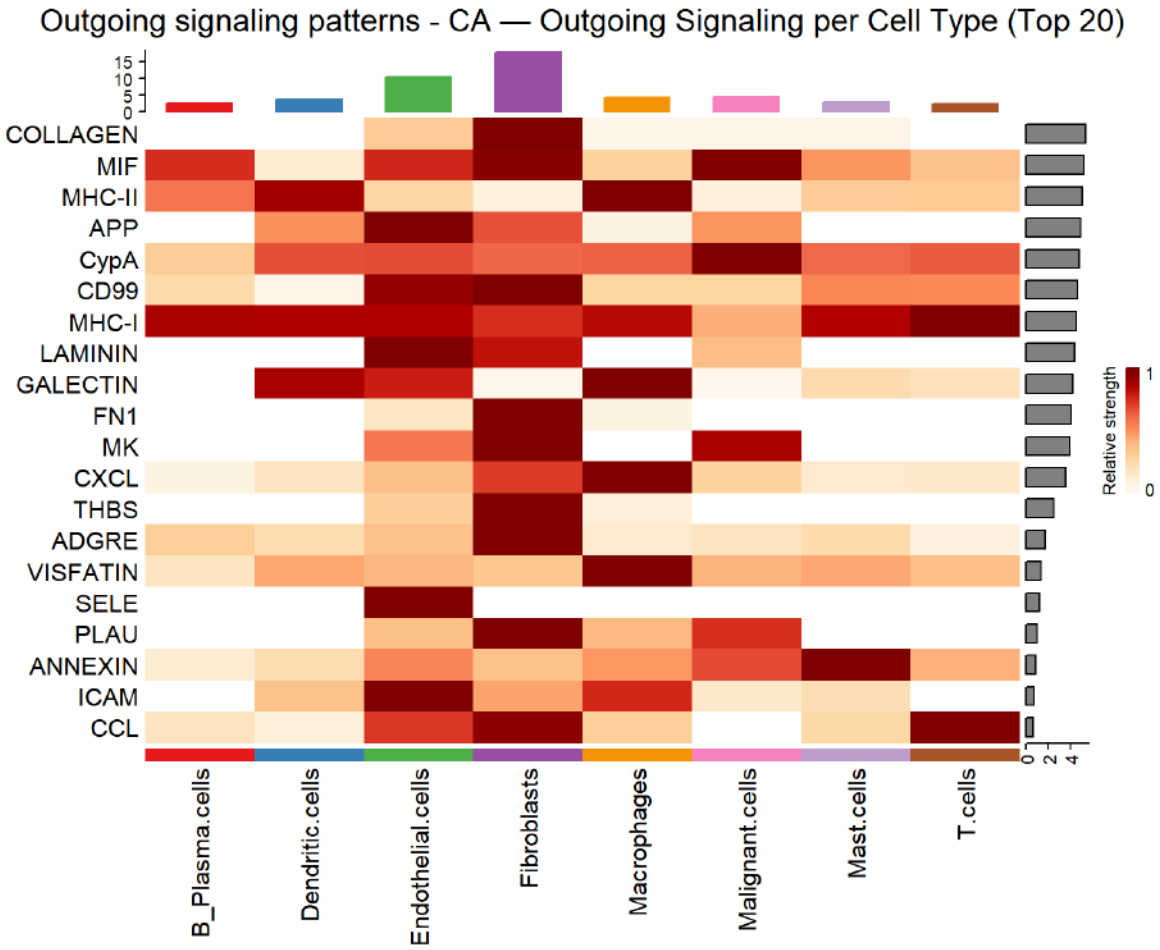

NL — Signaling Roles
T.cells
Macrophages
Dendritic.cells
Endothelial.cells
Fibroblasts
B_Plasma.cells
Mast.cells
Epithelial.cells
Myocytes
Incoming interaction strength
Outgoing interaction strength
0
1
2
3
4
5
0
2
4
6
Count
50
100
150
200

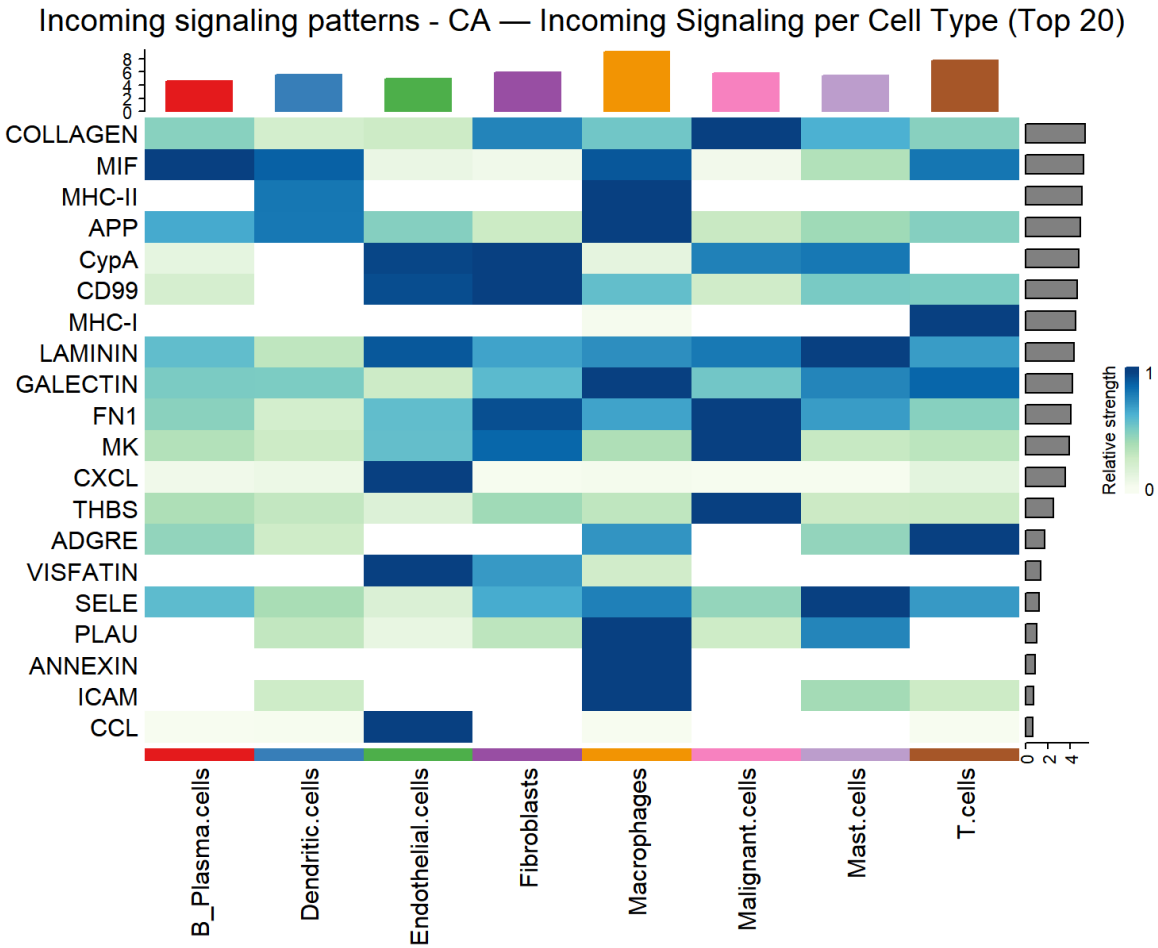

CA — Signaling Roles
Macrophages
T.cells
Malignant.cells
Dendritic.cells
Mast.cells
B_Plasma.cells
Endothelial.cells
Fibroblasts
Incoming interaction strength
Outgoing interaction strength
9
8
7
6
5
5
10
15
Count
300
400
500
600
700


LN — Signaling Roles
T.cells
Macrophages
Dendritic.cells
Malignant.cells
Endothelial.cells
Fibroblasts
Mast.cells
B_Plasma.cells
Incoming interaction strength
Outgoing interaction strength
4
6
8
10
12
5
10
15
Count
200
300
400
500
600


HPV– vs HPV+ — Differential Interactions
150
100
50
0
Sources (Sender)
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Malignant.cells
Mast.cells
T.cells
Relative values
20
0
-20
-40
-60

HPVneg vs HPVpos — Exhaustion Pathways → T Cells
LGALS9 - P4HB
LGALS9 - HAVCR2
LGALS9 - CD45
LGALS9 - CD44
HLA-F - CD8B
HLA-F - CD8A
HLA-E - CD8B
HLA-E - CD8A
HLA-DRB5 - CD4
HLA-DRB1 - CD4
HLA-DRA - CD4
HLA-DQB1 - CD4
HLA-DQA2 - CD4
HLA-DQA1 - CD4
HLA-DPB1 - CD4
HLA-DPA1 - CD4
HLA-DOB - CD4
HLA-DOA - CD4
HLA-DMB - CD4
HLA-DMA - CD4
HLA-C - CD8B
HLA-C - CD8A
HLA-B - CD8B
HLA-B - CD8A
HLA-A - CD8B
HLA-A - CD8A
CXCL9 - CXCR3
CXCL16 - CXCR6
CXCL13 - CXCR3
CXCL12 - CXCR4
CXCL10 - CXCR3
CD80 - CTLA4
CD80 - CD28
B_Plasma.cells -> T.cells (HPVneg)
B_Plasma.cells -> T.cells (HPVpos)
Dendritic.cells -> T.cells (HPVneg)
Dendritic.cells -> T.cells (HPVpos)
Endothelial.cells -> T.cells (HPVneg)
Endothelial.cells -> T.cells (HPVpos)
Fibroblasts -> T.cells (HPVneg)
Fibroblasts -> T.cells (HPVpos)
Macrophages -> T.cells (HPVneg)
Macrophages -> T.cells (HPVpos)
Malignant.cells -> T.cells (HPVneg)
Malignant.cells -> T.cells (HPVpos)
Mast.cells -> T.cells (HPVneg)
Mast.cells -> T.cells (HPVpos)
T.cells -> T.cells (HPVneg)
T.cells -> T.cells (HPVpos)
p-value
0.01 < p < 0.05
p < 0.01
Commun. Prob.
max
min


Outgoing signaling patterns - CA — Outgoing Signaling per Cell Type (Top 20)
15
10
5
0
COLLAGEN
MIF
MHC-II
APP
CypA
CD99
MHC-I
LAMININ
GALECTIN
FN1
MK
CXCL
THBS
ADGRE
VISFATIN
SELE
PLAU
ANNEXIN
ICAM
CCL
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Malignant.cells
Mast.cells
T.cells
0
2
4
Relative strength
1
0


Incoming signaling patterns - CA — Incoming Signaling per Cell Type (Top 20)
COLLAGEN
MIF
MHC-II
APP
CypA
CD99
MHC-I
LAMININ
GALECTIN
FN1
MK
CXCL
THBS
ADGRE
VISFATIN
SELE
PLAU
ANNEXIN
ICAM
CCL
B_Plasma.cells
Dendritic.cells
Endothelial.cells
Fibroblasts
Macrophages
Malignant.cells
Mast.cells
T.cells
Relative strength
1
0